

IntelliScore AI

Automated Essay Evaluation, AI Diagnostics & Writing Intelligence Platform

Version 2.4.0

FastAPI + React 18

Google Gemini 1.5/2.0 Flash

FAISS Vector RAG

Production Ready

1. Executive Abstract

IntelliScore AI is an enterprise-grade, full-stack educational technology platform engineered to automate multi-criteria essay scoring, detect AI-generated writing, identify plagiarism, and deliver actionable grammatical diagnostics. By fusing Large Language Models (LLMs) with high-speed Natural Language Processing (NLP) heuristic algorithms, IntelliScore AI reduces educator grading workloads by over 80% while providing students with instant, objective writing feedback.

2. System Architecture & Core Technologies

Frontend Stack

- **React 18** Component-driven UI framework
- **TypeScript** Type-safe client architecture
- **Vite** Modern build system & HMR
- **Tailwind CSS** Responsive Glassmorphism design system
- **Framer Motion** Dynamic micro-animations & transitions
- **jsPDF & DOCX** Executive PDF & Word report export engines

Backend & AI Stack

- **Python 3.11** Asynchronous backend core
- **FastAPI** Asynchronous ASGI web framework
- **SQLAlchemy** Relational ORM mapping
- **SQLite** Embedded persistent data store
- **Google Gemini LLM** 1.5/2.0 Flash evaluation engine
- **FAISS Vector Store** High-dimensional similarity vector database

3. AI & NLP Models Breakdown

Model / Component	Technology Provider	Functional Application
Primary Evaluation LLM	Google Gemini API (1.5 / 2.0 Flash)	Scores essays across 5 criteria (Grammar, Vocabulary, Structure, Readability, Argumentation), generates contextual line-by-line rewrites, and powers the RAG AI Writing Mentor.
AI Content Detector	Perplexity & Burstiness NLP Scoring	Analyzes sentence length variance, syntax predictability, and LLM token probability distributions to calculate exact AI vs. Human probability percentages.
Plagiarism & Similarity Engine	Cosine Similarity & TF-IDF Vector Space	Computes mathematical similarity matrices between submitted essays to detect text copying and inter-student plagiarism.
Readability Analyzer	Flesch-Kincaid & Gunning Fog Algorithms	Evaluates syllable counts, word densities, and sentence length ratios to determine grade-level readability scores.
RAG Vector Engine	Meta FAISS ('faiss-cpu')	Indexes essay embeddings for instant context retrieval during interactive AI Writing Mentor sessions.

4. Key Application Capabilities

- **Dynamic 5-Criteria Essay Evaluation:** Instant overall scoring out of 100 with sub-score breakdowns for Grammar (25%), Vocabulary (20%), Structure (20%), Readability (15%), and Argumentation (20%).
- **Grammar Diagnostic & Rewrite Suggestions:** Detects passive voice, subject-verb mismatches, and syntax errors, offering 1-click apply rewrite recommendations.
- **Role-Based Access & Strict Data Isolation:** Dedicated Student, Teacher, and Administrator roles. New teacher accounts receive clean, isolated workspaces (0 submissions), preventing cross-account demo data leaks.
- **Comprehensive Export Engines:** One-click generation of native Microsoft Word (`.docx`) and high-fidelity PDF evaluation reports.
- **Interactive AI Writing Mentor:** Conversational assistant that analyzes essay drafts and offers real-time guidance grounded in student writing context.

5. Database Schema Architecture

Entity / Table	Key Attributes	Description
users	id, email, full_name, role, hashed_password, institution, department	Stores registered users with role permissions (student, teacher, admin).
essays	id, title, original_text, file_type, user_id, category, created_at	Stores uploaded essay drafts and student metadata.
evaluations	id, essay_id, overall_score, grammar_score, vocabulary_score, ai_probability	Persists deep NLP analytics, category scores, and LLM feedback.

6. Deployment & Execution Instructions

Starting Backend Server (FastAPI)

```
cd backend && ..\.venv\Scripts\python.exe -m uvicorn app.main:app --reload --port 8000
```

Starting Frontend Server (React + Vite)

```
cd frontend/react-app && npm run dev
```